

The mixed fleet vehicle routing problem with dual load and environmental preferences of customers

Justin Wittig^a, Christian Bierwirth^a, Thomas Kirschstein^b

^a*Martin Luther University Halle-Wittenberg, School of Business and Economics, Chair of Production & Logistics, Halle (Saale), Germany*

^b*RheinMain University of Applied Sciences, Logistics & Supply Chain Management Group, Wiesbaden, Germany*

Abstract

Environmental regulations and the increasing relevance of corporate sustainability are accelerating the transition toward low-emission technologies in freight transportation. Among others, the transformation leads to heterogeneous vehicle fleets composed of conventional diesel trucks and battery-electric vehicles. Furthermore, shippers are beginning to procure transport services based on environmental considerations, creating novel challenges for tactical route planning. To address these developments, we propose an extension of the Mixed Fleet Vehicle Routing Problem (MFVRP) which incorporates individual customer preferences regarding cost efficiency and environmental performance. The resulting model, called MFVRP with maximizing the mean overall customer satisfaction (MFVRP-MS), is compared with conventional models minimizing total cost or transport emission. The approach is embedded in a full-cost accounting framework, grounded in current data on heavy-duty vehicles. We examine the extent to which customer preferences influence the structure of vehicle routes in a comprehensive computational study. The results demonstrate that conventional approaches frequently fail to accommodate individual customer priorities. In contrast, the MFVRP-MS consistently achieves a robust trade-off between economic and ecological objectives, especially in scenarios with partial fleet electrification and heterogeneous customer preferences.

Keywords: Electric truck cost analysis, mixed fleet vehicle routing, cost and emission allocation, customer preferences and satisfaction

1. Introduction

Stricter environmental requirements for road transportation and a growing demand for environmentally friendly services are causing carriers to gradually convert their fleets to low-emission and, in the long term, climate-neutral drive technologies. Vehicle fleets that currently consist of conventional diesel-powered trucks will be increasingly supplemented or replaced by trucks powered by synthetic fuels, fuel cells or battery-electric drives. This transformation process will extend over a long and as yet unforeseeable period of time. It can already be observed that the heterogeneity of drive types is increasing in commercial vehicle fleets (Ghandriz et al., 2021).

At the same time, incentives are provided to encourage industrial shippers choosing climate-friendly transport options. For example, upcoming standards for sustainability reporting require indirect emission caused by externally provided logistics services to be included in the product carbon footprint (Olivari et al., 2024). Industries operating in eco-sensitive markets could therefore increasingly request cleaner transport services. For instance, LEGO Group (2024) indicates that approximately 99% of its total emission is indirect emission, with 14% stemming from freight transportation alone. Such companies have a slightly larger willingness to pay for a transport service. In return they expect their shipments to be transported by low-emission vehicles, which carriers must prove by a certified CO₂ emission declaration (Wild, 2021). At the same time, cost-sensitive shippers still demand low transport prices and are prepared to accept higher emission declarations.

When heterogeneous preferences of shippers meet a carrier with a mixed vehicle fleet, a structurally novel planning problem arises. Route planning usually aims at economic efficiency and punctuality in customer deliveries, whereby it does not matter which vehicle serves which customer. However, if each vehicle use is billed individually according to valid standards, the assignment decisions of customer orders to vehicles impact customer satisfaction. Thus its likely that the technology mix of vehicle fleets will be more influenced by customer demand than is currently the case.

The paper is organized as follows: Section 2 provides a comprehensive review of literature covering heterogeneous vehicle routing, environmental preference modeling in transport planning, and cost controlling of heavy-duty vehicle fleets. Section 3 provides an informal description of the MFVRP-MS, the way vehicle characteristics and cost and emission allocation are handled, and, finally, the determination of customer satisfaction. In section 4, we

38 present a mixed-integer formulation of the MFVRP-MS based on a set par-
39 titioning approach, which is easily solved by Gurobi. Section 5 outlines the
40 experimental design, data sources, and the pre-processing used for scenario
41 generation. The computational study is divided into three parts. First, we
42 compare the structure of the vehicle routes generated by the MFVRP-MS
43 with the route structure obtained by minimizing total cost or total transport
44 emission. Then we study whether these routes are economically and ecologi-
45 cally viable. Finally, we examine the impact of maximizing either the mean or
46 the least level of customer satisfaction in the vehicle routing problem. The
47 results provide managerial insight into how carriers can align operational
48 routing decisions with economic and environmental goals by incorporating
49 customer preferences. The paper closes with an outlook on future research.

50 2. Previous research

51 In 2.1, we consider the development of vehicle routing models incorporating
52 heterogeneous fleets, with a particular focus on the integration of electric
53 trucks. In 2.2 we discuss regulatory and methodological developments re-
54 lated to the ecological assessment of transport operations. Finally, 2.3 re-
55 views Total-Cost-of-Ownership approaches for heavy-duty vehicles in order
56 to analyse the current competitive situation of electric trucks.

57 2.1. *Heterogeneous vehicle routing*

58 Since the late 1950s, the vehicle routing problem (VRP) is intensively studied
59 as a central component for tactical and operational planning of road haulage
60 companies (Laporte, 2009; Toth and Vigo, 2014). In the 1980s, route plan-
61 ning of heterogeneous fleets developed as a subfield of the VRP literature
62 (Koc et al., 2016). Triggered by Golden et al. (1984), model extensions were
63 developed and investigated under the name "fleet size and mix vehicle rout-
64 ing problem" (FSMVRP) to link decisions on fleet size and vehicle mix with
65 route planning. While the initial aim was to find an optimal mix of vehi-
66 cles with different fixed and variable operating costs, further problems were
67 later addressed in which vehicles also differ in terms of transport capacity,
68 equipment, speed and operating time (Lin et al., 2014).

69 For this research, VRP variants are of particular interest which take the
70 vehicle class and vehicle type into consideration. The vehicle class of a com-
71 mercial vehicle is defined by its gross weight, which is made up of its empty
72 weight and its payload. A common division is between light, medium and

heavy-duty vehicles (Koc et al., 2016). The vehicle type refers to the powertrain of the vehicle distinguishing between vehicles with a conventional internal combustion engine (ICEV) and vehicles with an electric drive (EV) (Pelletier et al., 2016). The latter is further subdivided into battery electric vehicles, fuel cell/hydrogen electric vehicles and hybrid electric vehicles. We refer to ICEVs as conventional trucks (c-trucks) in the following. The different types of EVs are collectively referred to as e-trucks.

It should be noted that e-trucks do not operate emission-free as long as electricity is generated by coal-fired power plants or synthetic fuels are not produced with renewable energy. To compare vehicles in terms of CO₂ equivalent emission, well-to-wheel (WTW) analysis of the vehicle systems is required (Brinkman et al., 2005; Gialos, 2022). In practice, the emission volume produced by c-trucks is determined based on fuel consumption and a WTW conversion factor. For diesel engines a value of 3.24 kg CO_{2e} per liter fuel is often assigned (Kranke et al., 2011). For e-trucks, the electricity consumption in kilowatt hours (kWh) is used accordingly. For the electricity mix in Germany, Icha and Lauf (2025) specify the current conversion factor at 0.434 kg CO_{2e} per kWh. A comparative WTW analysis of commercial vehicle classes is provided by the European Commission (2020).

From the large body of papers on mixed-fleet VRPs, we highlight three that are related to our research. Kopfer et al. (2014) demonstrate that the emission of a homogeneous fleet of c-trucks hardly decreases if vehicle routing is based on emission instead of traveling distance. This changes as soon as a heterogeneous fleet of c-trucks of different vehicle classes is available. Goeke and Schneider (2015) describe and solve a VRP with time windows for a mixed fleet of c-trucks and e-trucks. Integration of e-trucks can lead to cost reductions and emission savings depending on vehicle ranges and charging infrastructure. Hiermann et al. (2016) introduce the electric-FSMVRP with time windows and charging stations allowing simultaneous decisions on the design and routing of a fleet of e-trucks of different size. The allocation of transport emission to shippers and the associated pricing of services have not yet been considered in research on mixed fleet VRPs.

2.2. Ecological assessment of vehicle routing

The GHG Protocol and the ISO 14064 series of standards are accepted methods for accounting greenhouse gas (GHG) emission and determining a company’s carbon footprint. They cover emission caused by the company (scope 1), indirect emission caused by electricity consumption (scope 2) and further

indirect emission associated with the company’s activities (scope 3). Scope 3 emission is most difficult to report because it includes emission sources that the company does not control such as the use of external transport services. To ensure a transparent allocation of transport emission to shipments the European Norm EN 16258 has been developed (European Committee for Standardization, 2013). It contains, among others, rules how to allocate vehicle emission to customers in proportion to shipment characteristics such as payload, volume or distance traveled.

The volume of transport emission can be incorporated into vehicle routing in two ways. On the one hand it can be considered directly as a target to be minimized. On the other hand it can be used as a restriction to limit the emission shares of the costumers served in a process. The former type, also referred to as emission vehicle routing, has been extensively researched (Toth and Vigo, 2014; Demir et al., 2014). The latter type, which we focus on in this paper, assumes that customers have individual ecological preferences. There are relatively few papers that deal with the operational planning of transport processes taking into account emission limits of shippers.

Kirschstein and Bierwirth (2018) present a method to determine the lower limits for the CO_2 declaration of transport orders according to EN 16258. Heinold and Meisel (2018) develop a model for routing transport orders in an intermodal road-rail network with individual emission limits of customers. Kellner (2022) examines the effects of allocation rules on the environmental impact of transport decisions. Zhang et al. (2022) propose a cooperation model for freight forwarders operating intermodal transport networks. The aim is to meet the expectation of customers, expressed by eco-labels, that a carrier must comply with. Heinold et al. (2023) consider a stochastic dynamic intermodal transport problem with environmental preferences of customers. In summary, existing approaches to freight transport planning that take customer preferences into account focus primarily on shifting road transport to more environmentally friendly modes of transport.

2.3. Comparative vehicle cost analysis

The Total Cost of Ownership (TCO) framework is widely recognized as a suitable method for evaluating long-term cost structures in freight transport. It provides a holistic view of the expenses incurred throughout a vehicle’s service life, including acquisition, depreciation, maintenance, fuel or energy consumption, insurance, taxation, and residual value (Roosen et al., 2015; Parviziomran and Bergqvist, 2023).

TCO study results vary considerably depending on the considered vehicle class. As emphasized by Danielis et al. (2025), light and medium-duty battery-electric trucks have started to become cost-competitive as early as 2021. However, heavy-duty trucks (HDVs) remain economically disadvantaged, and may continue to struggle reaching cost parity with diesel alternatives even in the coming decade. The cost gap is driven by factors such as battery prices, limited daily range, and longer charging times, all of which affect the cost-effectiveness of e-trucks in long-haul freight transportation.

While full TCO models aim to capture all direct and indirect cost components, many operational studies, including this one, adopt a more targeted cost accounting structure that distinguishes between fixed and variable cost elements. This differentiation enables analytically tractable and practically relevant assessments of transport cost, which are crucial for tactical planning and cost allocation in logistics operations. Fixed vehicle costs typically encompass time-based components such as depreciation, capital interest, insurance, and taxes. These costs are independent of the vehicle usage and commonly annualized based on the expected service life and utilization pattern of the vehicle. Variable vehicle costs, by contrast, scale with the operational intensity. They include distance-based expenses such as energy or fuel cost and tolls (Syré et al., 2025). This separation aligns with recent approaches in the literature, allowing for a transparent and scalable estimation of transport cost in fleet-based optimization models.

From a carrier’s perspective, the estimation of the cost per tour based on fixed and variable vehicle cost provides the foundation for deriving economically sound and customer-facing transport rates. Such rates are especially important in part-load operations, where transport costs must be allocated across multiple customers served on the same tour. Our study follows this rationale by developing cost rates on a per-day and per-kilometer basis, enabling the proportional allocation of a tours’ expenses to individual customers according to their location and share of the route.

3. Mixed fleet vehicle routing problem with customer preferences

This section sketches the MFVRP-MS. Following a brief problem description in 3.1, methods for calculating and allocating vehicle cost and emission are presented in 3.2. Our approach to determining customer satisfaction is introduced in 3.3. Finally, in 3.4, an example is presented to illustrate differences between MFVRP-MS solutions and common VRP variants.

183 3.1. Problem description and assumptions

184 We consider a VRP for long-haul road transportation with partial loads. The
185 carrier operates from a central depot with a heterogeneous fleet consisting
186 of identical c-trucks and identical e-trucks. Both truck types are from the
187 heavy-duty class of vehicles. They have a same capacity of two twenty-foot
188 equivalent units (TEU) but they can differ in terms of cost and emission.

189 A set of shippers, referred to as customers, place transport orders to
190 the carrier. Transport orders consist of a loading unit of one TEU and a
191 destination specified by the customer. The transport orders are handled
192 as part of a delivery process, in which a vehicle route starts at the depot,
193 serves a number of customers in line with the vehicle capacity, and ends at
194 the depot. We assume that the carrier is given all transport orders at the
195 beginning of a period and is capable executing them within that period. We
196 further assume that every customer communicates a preference regarding the
197 way its transport order should be fulfilled. This divides the set of customers
198 into cost-sensitive and emission-sensitive ones.

199 Once all data has been collected, the carrier solves the associated MFVRP
200 with the aim of maximizing customer satisfaction. With the solution at hand,
201 all customers receive a price quote and an expected volume of CO₂ emission
202 which both are in line with a transparent tariff system and a valid emission
203 allocation standard. Since customer preferences are taken into account when
204 solving the problem, we further assume that all customers accept the offered
205 service. The way prices, costs and emissions for offered transport services
206 are fixed and customer satisfaction is determined are described below.

207 3.2. Fixing prices, costs and emissions for transport services

208 In order to determine meaningful prices for conventional and environmentally
209 friendly road transport services, the vehicle cost are calculated on a modified
210 full-cost basis. As an example we refer to the *Mercedes-Benz Actros*, which is
211 offered with a conventional diesel combustion engine and with a battery elec-
212 tric drive. The purchase prices currently amount to €150,000 and €300,000,
213 respectively (Mercedes-Benz Trucks, 2025), see Table 1.

214 A linear depreciation method is used to account for the loss in value of
215 the vehicles during service life. According to Zhou et al. (2017) service life
216 is set to 9 years for a c-truck. For an e-truck, we reduce it to 6 years due
217 to battery degradation. Both types of trucks can be used on 240 days per
218 year. The annual depreciation rate corresponds to the difference between the
219 expected repurchase price and the residual value divided by the service life

Table 1: Vehicle cost calculation (modified full-cost basis)

Parameter	ActrosL	eActros 600 MB
<i>Vehicle data</i>		
Net vehicle price	150,000 €	300,000 €
Service life in years	9	6
Days of use per year	240	240
Residual value after service life	20,000 €	50,000 €
Expected repurchase price	140,000 €	250,000 €
Annual mileage	120,000 km	90,000 km
Energy consumption	33 l/100km	103 kWh/100km
<i>Economic variables (status Germany, 2025)</i>		
Interest rate	5 %	5 %
Energy price	1.50 €/l fuel	0.55 €/kWh
Road toll	1.85 €/100km	0.00 €/100km
<i>Cost calculation</i>		
Time-dependent depreciation p.a.	6,667 €	16,667 €
Imputed interest p.a.	3,750 €	7,500 €
Vehicle tax p.a.	1,000 €	[†] 0 €
Insurance p.a.	4,000 €	6,000 €
Driver wages p.a.	60,000 €	60,000 €
Total fixed cost p.a.	75,417 €	90,167 €
Operation-dependent depreciation p.a.	6,667 €	16,667 €
Energy cost p.a.	59,400 €	50,985 €
Road toll p.a.	2,220 €	[‡] 0 €
Total variable cost p.a.	68,287 €	67,652 €
<i>Summary of cost</i>		
Total vehicle cost p.a.	143,704 €	157,819 €
Cost rate per day c_{day}	314.24 €	375.69 €
Cost rate per kilometer c_{km}	0.57 €	0.75 €

[†] E-trucks are exempt from vehicle tax under §9(2) of the German vehicle tax law.

[‡] E-trucks are exempt from road toll (Federal Office for Logistics and Mobility, 2024).

span. It represents provisions for the replacement of the truck at the end of its service life. The expected repurchase prices and residual values are estimated as percentages of the current net value prices. We assume that half of the vehicle depreciation is based on time and half on kilometric performance. For the c-truck, this leads to an annual time-dependent and operation-dependent depreciation of €6,667 each. For the e-truck the rates amount to €16,667 each due to the higher purchase price. The annual mileage of a c-truck is expected to be about 120,000 km. For e-trucks we reduce it by 25% to take charging processes into consideration. The energy consumption rates shown in the table are based on the manufacturer’s specification.

Fixed annual vehicle cost correspond to the sum of time-dependent depreciation, imputed interest, and annual expenditures for vehicle tax, insurance and driver wages. The imputed interest reflects the capital tied up on average during the vehicle’s service life. It corresponds to an annual interest on half the net vehicle price at an interest rate, which is assumed to be 5%.

Variable annual vehicle cost correspond to the sum of operation-dependent depreciation, annual energy cost, and annual road toll. The latter expenditures are computed from the expected annual mileage of the trucks, their energy consumption, and estimated prices for energy and road toll. We disregard other factors such as cost of maintenance and tires because they are similar across the vehicle types and thus do not impact our study.

The shown energy prices correspond to average values observed in Germany in 2025. For simplicity we assume that the carrier does not have its own charging infrastructure. Therefore, the procurement and amortization of a suitable infrastructure at the depot has not to be taken into account. According to Bragin et al. (2023) and Shoman et al. (2023), public charging infrastructure for long-haul battery-electric trucks typically relies on combined charging systems (CCS) or megawatt charging systems (MCS). Milence (2024) is a European charging network, established as a joint venture between Daimler Truck, Volvo Group, and others. The company operates a CCS and currently demands €0.48 per kWh for public charging. The advanced MCS technology is not yet operational. In our calculation, we assume a future price of €0.55 per kWh in order to be prepared for expected price increases coming along with the switch to the faster MCS technology.

The total vehicle cost sums up to €143,704 for the ActosL. For the eActos, they are approximately 10% higher at €157,819. In order to determine the carrier’s service cost for a single tour, two cost rates are derived for each vehicle type. The cost per day c_{day}^k divides the annual fixed cost by the

Table 2: Vehicle emission calculation

Parameter	ActrosL	eActros 600 MB
Energy consumption	0.33 l/km	1.03 kWh/km
WTW emission factor	[†] 3.24 kg CO ₂ e/l	[‡] 0.43 kg CO ₂ e/kWh
Emission rate per kilometer e_{km}	1.07 kg CO ₂ e/km	0.44 kg CO ₂ e/km

[†] applies to conventional diesel fuel (McKinnon and Piecyk, 2010)

[‡] applies to electricity from the German grid (Icha and Lauf, 2025)

average days of use. It measures the expanses of using a truck of type k for a day without moving it. The cost per kilometer c_{km}^k divides the annual variable cost by the expected annual mileage. The carrier's cost for a tour with a vehicle of type k over a length of TL kilometer is estimated as $c_{day}^k + c_{km}^k \cdot TL$. It is distributed among all customers $i \in C$ served on the tour by

$$cost_i = \frac{1}{|C|} \cdot c_{day}^k + \frac{d_{0i}}{\sum_{j \in C} d_{0j}} \cdot c_{km}^k \cdot TL. \quad (1)$$

The fixed cost portion is shared equally among all customers while the variable cost portion is allocated in proportion of the respective allocation distance of customer i . It is defined as the direct driving distance from the depot to the customers destination. We further assume that the carrier charges an additional profit premium to each customer which we ignore in the following for the sake of simplification. Hence, freight prices correspond to the tour cost carried out under the described cost allocation scheme.

The CO₂e emission caused by a truck of type k during a tour is distributed according to allocation rules specified by EN16258. For simplicity, we choose distance-based allocation. It calculates the emission volume assigned to customer $i \in C$ as

$$emis_i = \frac{d_{0i}}{\sum_{j \in C} d_{0j}} \cdot e_{km}^k \cdot TL, \quad (2)$$

where e_{km}^k denotes the emission coefficient of vehicle type k . These values are shown in Table 2 for the ActrosL and eActros trucks. They specify the WTW emission that a truck produces when traveling a distance of one kilometer. It results by multiplying the energy consumption rate of the vehicle shown in Table 1 with the WTW conversion factor of the used energy source.

279 3.3. Evaluating customer satisfaction

280 A new target is introduced for the VRP, which we refer to as the optimiza-
 281 tion of relative customer satisfaction. It is defined by a two-dimensional
 282 metric that measures the economical and the ecological impact of the service
 283 provided to each single customer. Economic satisfaction is measured by the
 284 ratio of the cost to be paid by a customer for the projected tour against the
 285 minimum possible cost to be paid when the same customer would have been
 286 served within a vehicle tour being ideal for this customer.

287 Since a dual load VRP is considered in this paper, no more than two
 288 customers can be served together on a tour. The ideal partner of a customer
 289 is determined according to the well known saving approach. If customers i
 290 and j are served together, the tour length adds up to $TL = d_{0i} + d_{ij} + d_{0j}$.
 291 Each of the two customers pays the kilometer-based cost for its own allocation
 292 distance while the cost for leg d_{ij} is shared in proportion of the two allocation
 293 distances. Thus, sharing a tour with customer j is ideal for i if

$$s_{ij} = \frac{d_{0i}}{d_{0i} + d_{0j}} \cdot d_{ij} \quad (3)$$

294 takes its minimum for $j \in C \setminus \{i\}$. In other words, the ideal partner $j \in C$ of
 295 customer i is determined as customer $j^*(i) = \operatorname{argmax}_{j \in C \setminus \{i\}}(s_{ij})$.

296 Let $\underline{c}_i$ and $\bar{c}_i$ denote the lowest and highest cost chargeable to customer
 297 $i \in C$ for a given VRP instance. Cost $\underline{c}_i$ is charged if customer i and its ideal
 298 partner $j(i)^*$ share a service executed by a c-truck as this vehicle type has
 299 typically lower cost rates than an e-truck. On the other hand, $\bar{c}_i$ occurs if
 300 customer i is served with its least ideal partner using an e-truck. Given the
 301 carrier calculates the service price for the customers using equation (1), the
 302 economic relative satisfaction of customer i is evaluated as

$$CS_i = 1 - \frac{\text{cost}_i - \underline{c}_i}{\bar{c}_i - \underline{c}_i}. \quad (4)$$

303 If customer i is charged the highest or lowest possible cost, $CS_i = 0\%$
 304 or 100% , respectively. Note that $\underline{c}_i$ and $\bar{c}_i$ are independent of vehicle routes
 305 and, thus, can be pre-processed.

306 This concept can be transferred to ecological customer satisfaction. Let $\underline{e}_i$
 307 and $\bar{e}_i$ denote the lowest and highest emission volume allocatable to customer
 308 i . Volume $\underline{e}_i$ is allocated to i if the customer shares an e-truck service with
 309 an ideal partner while volume $\bar{e}_i$ is allocated if i is served together with the

310 least ideal partner by a c-truck. Following (4), the customer's satisfaction
311 about a declaration of $emis_i$ kg CO₂e is

$$ES_i = 1 - \frac{emis_i - \underline{e}_i}{\bar{e}_i - \underline{e}_i}. \quad (5)$$

312 Let each customer now express an individual environmental attitude,
313 where $pref_i = 1$ indicates an emission-sensitive and $pref_i = 0$ a cost-sensitive
314 customer. With its choice a customer can encourage the carrier to increase
315 its emission declaration in favour of lower transport cost, or vice versa. The
316 carrier determines the overall satisfaction of customer i by

$$OS_i = pref_i \cdot ES_i + (1 - pref_i) \cdot CS_i. \quad (6)$$

317 The measure consists of two percentages from which the customer selects one.
318 Basically, customers could also divide their weight upon both percentages
319 and, in this way, express a more or less indifferent attitude. Likewise, the
320 approach is also applicable when more than two customers can be served on
321 a route. However, we do not consider this possibility in this paper.

322 3.4. Illustrative MFVRP-MS example

323 To illustrate potential effects of satisfaction-oriented vehicle routing, we con-
324 sider a carrier with a fleet consisting of one c-truck and one e-truck. The
325 capacity of both vehicles is two TEU each. Cost and emission rates are given
326 as specified in Table 1 and Table 2. The carrier has received four trans-
327 port orders of one TEU each. Figure 1a shows the four destinations A-D
328 with the vehicle depot located at (0,0) in a Euclidean grid. It is assumed

Table 3: Total cost, emission, and their allocation to customers (lowest values are bold).

Plan	Total cost	Total emission	A cost	B emission	C emission	D cost
1	€ 856.59	271.84 kg	€ 228.57	119.09 kg	9.32 kg	€ 203.73
2	€ 891.56	149.45 kg	€ 281.85	48.97 kg	22.66 kg	€ 169.19
3	€ 888.88	235.55 kg	€ 236.84	53.37 kg	23.20 kg	€ 203.73
4	€ 892.31	223.54 kg	€ 292.75	129.79 kg	9.54 kg	€ 169.19
5	€ 869.17	211.32 kg	€ 228.34	48.43 kg	8.46 kg	€ 168.16
6	€ 871.87	201.85 kg	€ 281.56	117.78 kg	20.57 kg	€ 202.37

329 that customers A and D are emission-sensitive, and customers B and C are
 330 cost-sensitive. The problem is solved by complete enumeration. There are
 331 three different vehicle routes, each with two options for assigning trucks.
 332 Figure 1b-d shows the resulting six dual load tour plans.

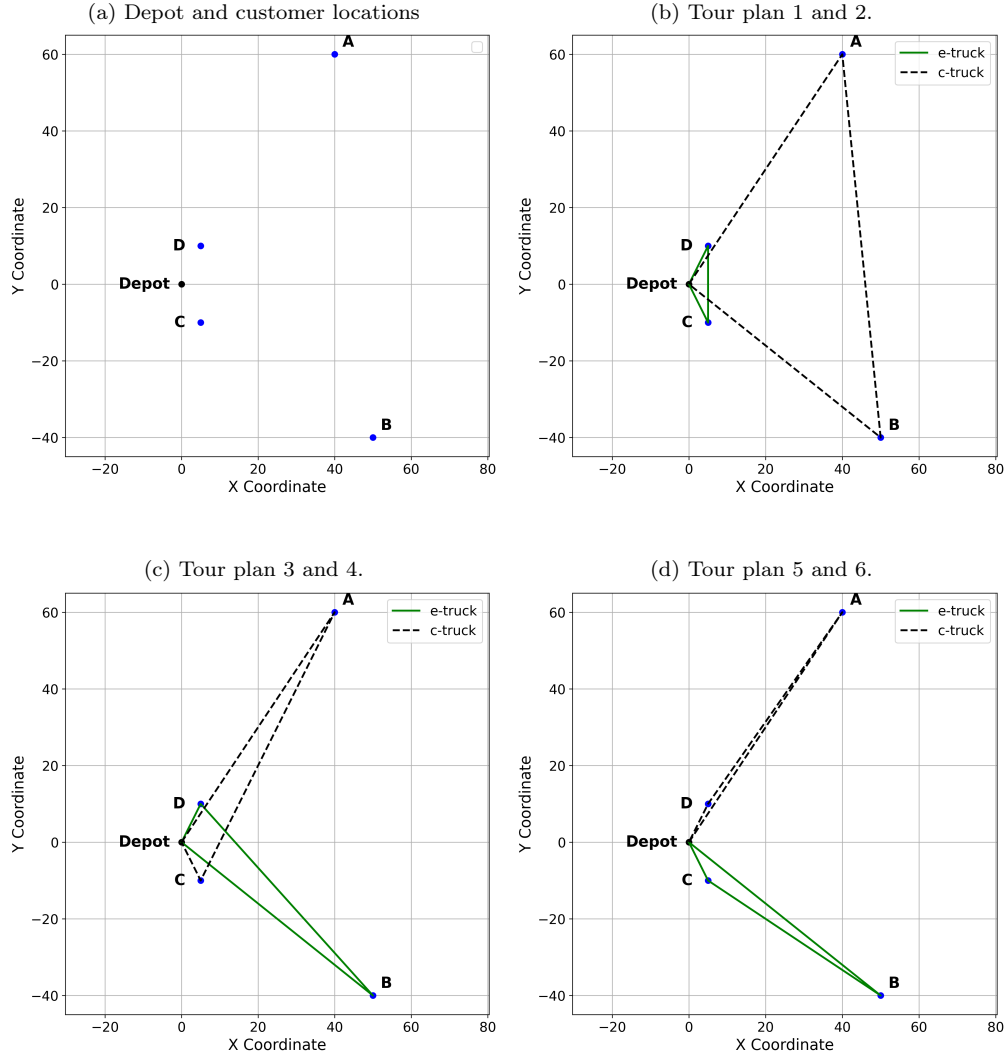


Figure 1: Problem and solutions in the 4-customer example.

333 The total cost and emission of the six tour plans are computed as ex-

334 plained in 3.2 and shown in Table 3. Plan 1 is optimal if total cost is mini-
 335 mized while Plan 2 is optimal if total emission is minimized. In Plan 1 (see
 336 Figure 1b), the c-truck serves distant customers A and D while the e-truck
 337 serves closer customers B and C. Plan 2 reverses this allocation.

338 The allocation of cost and emission according to (1) and (2) are shown
 339 in columns A to D of Table 3. The results indicate that neither the cost-
 340 sensitive customers nor the emission-sensitive customers will be completely
 341 satisfied with either Plan 1 or Plan 2. Notably, Plan 5 is the most satisfactory
 342 service for all customers, but total cost and total emissions increase by 1.5%
 343 and 41% above minimum, respectively. It should be noted that such solu-
 344 tions do not exist for every MVRP-MS instance. The example demonstrates
 345 that satisfaction-oriented vehicle routing can result in structurally different
 346 solutions compared to traditional VRPs.

347 4. Model formulations

348 This section provides MIP formulations for several variants of the MFVRP.
 349 The basic model captures the standard routing logic for a mixed fleet of ve-
 350 hicles with load capacity restrictions. It is used to minimize travel distance,
 351 cost, or total emission. It is then expanded to include customer preferences
 352 and two objective functions for measuring customer satisfaction. The result-
 353 ing models are finally reformulated as set partitioning problems in order to
 354 speed up the MIP solving process.

355 4.1. Basic vehicle routing models

356 Let N be a set of n customer nodes, N_0 the set including the depot node, and
 357 d_{ij} the traveling distances between any two nodes $i, j \in C_0$. Furthermore, let
 358 F be a set of vehicles representing a mixed fleet. In a dual load transportation
 359 process, the load weight of the customers is half of the vehicle's payload
 360 capacity. Therefore, a maximum of two customers can be visited during each
 361 vehicle tour. It is assumed that the size of fleet F is greater than $n/2$, so that
 362 no vehicle needs to be used twice to serve customers, i.e. $|F| = v \geq n/2$.
 363 However, the mix of the fleet, i.e. $v = v_c + v_e$, is variable, meaning that
 364 depending on the objectives pursued, either the number of c-trucks (v_c) or
 365 e-trucks (v_e) might be scarce.

366 A three-index based formulation is used to model the vehicle routing
 367 problem of the carrier. Binary decision variables x_{ij}^k indicate whether vehicle
 368 $k \in F$ travels directly from node i to node j or not. The fleet size and

369 mix do not appear explicitly as parameters in the model. Instead, vehicle
370 characteristics such as payload capacity, cost, and emission rates are given
371 individually for each element of F . This allows for the inclusion of any
372 heterogeneous fleet consisting of vehicles of different drive types and size
373 classes. In our computations, only vehicle cost and emission rates are used to
374 distinguish between e-trucks and c-trucks, see Table 7. For a generated tour
375 plan, the parameters are used to determine the tour cost TC^k and emission
376 TE^k incurred. Table 4 shows the notation of the MFVRP base models.

Table 4: MFVRP base model notation

Sets	
N	set of customer nodes, $N = \{1, 2, \dots, n\}$
N_0	set of customer nodes including the depot, $N_0 = N \cup \{0\}$
T	set of vehicle types, $T = \{c, e\}$
$F = \bigcup_t F_t$	set of all vehicles (fleet) of size $v = \sum_t v_t$ with $v_t = F_t $
Parameters	
d_{ij}	Euclidean distance between costumer nodes $i, j \in N_0$, (in km)
q_i	load weight of customer $i \in N$, (in TEU)
Q^k	payload capacity of vehicle $k \in F$, (in TEU)
c_{day}^k	vehicle cost rate per day of vehicle $k \in F$, (in €)
c_{km}^k	vehicle cost rate per kilometer of vehicle $k \in F$, (in €/km)
e_{km}^k	emission rate per kilometer of vehicle $k \in F$, (in kg CO ₂ e/km)
Decision variables	
x_{ij}^k	binary (1=vehicle $k \in F$ travels from $i \in N_0$ to $j \in N_0$, 0=else)
Decision-dependent variables	
y^k	binary, deployment of vehicle $k \in F$ (1=vehicle is deployed, 0=else)
u_i^k	load weight of vehicle $k \in F$ after visiting customer $i \in N$ (in TEU)
TL^k	tour length of vehicle $k \in F$ (in km)
TC^k	tour cost incurred by vehicle $k \in F$ (in €)
TE^k	tour emission caused by vehicle $k \in F$ (in kg CO ₂ e)

377 The first model, referred to as MFVRP-TD, aims to minimize the total

378 distance TL^k traveled by the vehicle fleet to serve customer transport orders
 379 as follows:

$$\min \rightarrow \sum_{k \in F} TL^k = \sum_{k \in F} \sum_{i \in N_0} \sum_{\substack{j \in N_0 \\ i \neq j}} d_{ij} x_{ij}^k \quad (7)$$

$$\text{s.t. } \sum_{k \in F} \sum_{\substack{i \in N_0 \\ i \neq j}} x_{ij}^k = 1 \quad \forall j \in N \quad (8)$$

$$\sum_{\substack{i \in N_0 \\ i \neq j}} x_{ij}^k = \sum_{\substack{i \in N_0 \\ i \neq j}} x_{ji}^k \quad \forall k \in F, \forall j \in N \quad (9)$$

$$y^k = \sum_{j \in N} x_{0j}^k \quad \forall k \in F \quad (10)$$

$$TL^k = \sum_{i \in N_0} \sum_{\substack{j \in N_0 \\ i \neq j}} d_{ij} x_{ij}^k \quad \forall k \in F \quad (11)$$

$$u_i^k - u_j^k + Q^k x_{ij}^k \leq Q^k - q_j \quad \forall k \in F, \forall i, j \in N_0, i \neq j \quad (12)$$

$$q_i \leq u_i^k \leq Q^k \quad \forall k \in F, \forall i \in N_0 \quad (13)$$

$$y^k, x_{ij}^k \in \{0, 1\} \quad \forall k \in F, \forall i, j \in N_0, i \neq j \quad (14)$$

$$u_i^k \geq 0 \quad \forall k \in F, \forall i \in N_0 \quad (15)$$

380 The objective function (7) minimizes the total distance traveled by the fleet.
 381 Constraint (8) ensures that each customer is visited exactly once by any
 382 vehicle. Flow conservation is maintained by (9). Equation (10) calculates
 383 the frequency of a vehicle's deployment, and (??) limits this to a maximum
 384 of one. The tour length is determined for the vehicles in (11). For unde-
 385 ployed vehicles, this results in a distance of zero kilometer. Together, (12)
 386 and (13) ensure that the vehicle capacity is not exceeded and that sub-tours
 387 are eliminated using the Miller-Tucker-Zemlin formulation. Binary and non-
 388 negativity conditions are specified in (14) and (15), respectively.

389 Building upon the transport cost and emission framework presented in
 390 3.2, we introduce two further models. The MFVRP-TC aims at minimizing
 391 the total operational cost of the carrier, while the MFVRP-TE aims at mini-
 392 mizing the total emission produced by the vehicle fleet to serve the customer
 393 transport orders. Both models result from a minor change of the objective

394 function in the MFVRP-TD as follows:

$$\min \rightarrow \sum_{k \in F} TC^k \quad \text{s.t.} \quad (8)-(15) \quad (16)$$

$$TC^k = c_{day}^k \cdot y^k + c_{km}^k \cdot TL^k \quad \forall k \in F \quad (17)$$

$$\min \rightarrow \sum_{k \in F} TE^k \quad \text{s.t.} \quad (8)-(15) \quad (18)$$

$$TE^k = e_{km}^k \cdot TL^k \quad \forall k \in F \quad (19)$$

395 Due to the daily vehicle cost rate in (17), model MFVRP-TC can produce
 396 optimal solutions different than those of model MFVRP-TD. In contrast,
 397 MFVRP-TD and MFVRP-TE are equivalent as the objective functions (7)
 398 and (18) are proportional to each other. Without loss of generality, MFVRP-
 399 TD is omitted from further discussion.

400 4.2. Models for optimizing customer satisfaction

401 The basic model is expanded with a view to optimizing the fulfillment of
 402 customer preferences. Additional notation is described in Table 5.

Table 5: Additional notation for MFVRP-MS and MFVRP-LS

Parameters	
$pref_i$	preference of customer $i \in N$, (1= low emission, 0= low cost)
$\underline{c}_i$	least cost chargeable from customer $i \in N$ (in €)
$\bar{c}_i$	most cost chargeable from customer $i \in N$ (in €)
$\underline{e}_i$	least emission volume allocatable to customer $i \in N$ (in kg CO ₂ e)
$\bar{e}_i$	most emission volume allocatable to customer $i \in N$ (in kg CO ₂ e)
Decision-dependent variables	
z_i^k	binary (1=customer $i \in N$ is served by vehicle $k \in F$, 0=else)
m^k	number of customers served by vehicle $k \in F$
s_i	allocation share of customer $i \in N$ (in %)
$cost_i$	cost allocated to customer $i \in N$, (in €)
$emis_i$	emissions allocated to customer $i \in N$ (in kg CO ₂ e)
CS_i	cost satisfaction of customer $i \in N$ (in %)
ES_i	environmental satisfaction of customer $i \in N$ (in %)
OS_i	overall satisfaction of customer $i \in N$ (in %)
LS	least overall satisfaction of any of the customers (in %)

403 The resulting MIP model, referred to as MFVRP-MS, is defined as fol-
404 lows:

$$\max \rightarrow \frac{1}{n} \sum_{i \in N} OS_i \quad \text{s.t.} \quad (8)-(15), \text{ and } (19) \quad (20)$$

$$z_i^k = \sum_{j \in N_0} x_{ji}^k \quad \forall k \in F, \forall i \in N \quad (21)$$

$$m^k = \sum_{i \in N} z_i^k \quad \forall k \in F \quad (22)$$

$$s_i = \sum_{k \in F} \frac{z_i^k \cdot d_{0i}}{\sum_{j \in N} z_j^k \cdot d_{0,j}} \quad \forall i \in N \quad (23)$$

$$cost_i = \sum_{k \in F} z_i^k \left(\frac{c_{km}^k}{m^k} + s_i \cdot c_{km}^k \cdot TL^k \right) \quad \forall i \in N \quad (24)$$

$$emis_i = \sum_{k \in F} z_i^k \cdot s_i \cdot TE^k \quad \forall i \in N, \quad (25)$$

$$CS_i = 1 - \frac{cost_i - \underline{c}_i}{\bar{c}_i - \underline{c}_i} \quad \forall i \in N \quad (26)$$

$$ES_i = 1 - \frac{emis_i - \underline{e}_i}{\bar{e}_i - \underline{e}_i} \quad \forall i \in N \quad (27)$$

$$OS_i = pref_i \cdot ES_i + (1 - pref_i) \cdot CS_i \quad \forall i \in N \quad (28)$$

$$z_i^k \in \{0, 1\} \quad \forall k \in F, \forall i \in N \quad (29)$$

$$m^k, s_i \geq 0 \quad \forall k \in F, \forall i \in N \quad (30)$$

405 The objective function (20) maximizes the mean overall satisfaction of the
 406 customers. Variables z_i^k in (21) are assigned, indicating whether customer
 407 i is visited by vehicle k or not. The number of customers m^k assigned to
 408 vehicle k is determined in (22). Based on these decision-dependent variables,
 409 the overall satisfaction OS_i of customer i is calculated through constraints
 410 (23)-(28) which resemble (1)-(6) introduced in subsection 3.3. Constraints
 411 (29) and (30) specify binary and non-negativity conditions required for the
 412 decision-dependent variables added to the MFVRP-MS.

413 Another variant of the MFVRP pursues maximizing the satisfaction of
 414 the most dissatisfied customer. With this objective, a carrier may attempt
 415 to prevent the majority of customers from receiving preferential treatment
 416 at the expense of a few others. The MFVRP-LS is defined as follows:

$$\max \rightarrow LS \quad \text{s.t.} \quad (8)-(15), (19) \text{ and } (21)-(30) \quad (31)$$

$$LS \leq OS_i \quad \forall i \in N \quad (32)$$

417 The MFVRP-LS strives for a leveling target widespread in fields of op-
 418 timization where fairness is viewed crucial. As the satisfaction of the most
 419 dissatisfied customer increases, it is hoped that the perceived differences in
 420 customer treatment disappear and, at the same time, that the average satis-
 421 faction level rises. However, reformulating a max-min objective by the linear
 422 constraints (32) increases complexity. Likewise, linearizing (24) and (25) is
 423 straight forward but increases the size of the model substantially. Therefore,
 424 the MILPs are reformulated as set partitioning models which seem to be
 425 more efficient for the problems at hand.

4.3. Set partitioning formulation

In the special case of dual-load transportation, where each vehicle serves at most two customers per tour, the MFVRP admits an alternative interpretation as an assignment problem. By partitioning the customer set N into two disjoint subsets N_1 and N_2 , any feasible tour can be represented as a pairing (i, j) with $i \in N_1$ and $j \in N_2$, operated by a vehicle of type $t \in T$. Each such assignment corresponds to a depot– i – j –depot tour, whose cost, emissions, and resulting customer satisfaction levels can be computed in a preprocessing step.

Under this perspective, the problem can be formulated as a weighted bipartite matching problem with parallel edges representing different vehicle types. If vehicle availability is unrestricted, the optimal vehicle type can be selected independently for each customer pair, and the resulting assignment problem can be solved in polynomial time using the Hungarian algorithm. Similar reductions apply for classical objective functions such as minimizing total distance, cost, or emissions.

Despite this appealing structure, the assignment-based formulation is inherently limited to the dual-load setting and does not extend naturally to more general routing problems with higher vehicle capacities or longer customer sequences. Moreover, incorporating additional routing constraints or richer fleet heterogeneity would substantially reduce its tractability. Therefore, in order to maintain a unified modeling and solution framework that remains applicable beyond the dual-load case, we adopt a set partitioning formulation in the following. This approach preserves the structural insights of the assignment perspective while allowing for a direct generalization to more realistic MFVRP variants.

Given a set of n customer orders, the number of different vehicle routes, starting at the depot, visiting exactly two of the customers and returning to the depot, is $\binom{n}{2}$. Since every route can either be served by a c-truck or an e-truck, exactly $\frac{n!}{(n-2)!}$ routes need to be distinguished in the n -customer MFVRP. For simplicity, we assume that n is even such that no routes have to be considered in which only a single customer is visited. We compile the set of possible vehicle routes that are meaningfully distinguished and denote it as R .

Having compiled set R , the route-dependent performance measures can be computed in a pre-processing step. For every $r \in R$ the visited customers a_r^i and serving vehicle type b_r^t are indicated. Then, total cost and emission

Table 6: Model notation for the set partitioning formulation of the MFVRP variants

Sets	
N	set of customers $N = \{1, 2, \dots, n\}$
T	set of vehicle types available in the fleet (here: 1=c-truck, 2=e-truck)
R	set of vehicle routes (taking the vehicle type into account))
Parameters	
v_t	number of vehicles of type $t \in T$ belonging to the fleet
a_r^i	binary (=1 indicates that customer $i \in N$ is served on route $r \in R$)
b_r^t	binary (=1 indicates that vehicle type $t \in T$ drives route $r \in R$)
VC_r	vehicle cost incurred on route $r \in R$. (in €)
VE_r	vehicle emission produced on route $r \in R$ (in kg CO ₂ e)
MS_r	mean overall satisfaction of customers served on route $r \in R$ (in %)
LS_r	least overall satisfaction of a customer served on route $r \in R$ (in %)
Decision variables	
x_r	binary (1=route $r \in R$ is included in the tour plan, 0=else)
Decision-dependent variables	
LS	least overall satisfaction of any of the customers (in %)

are calculated depending on the type of vehicle in use. Moreover, the satisfaction of each customer is determined when served on route r . That is why vehicle cost and emission rates, customer preference weights and further parameters listed in Table 4 and Table 5 disappear in the reformulation of the optimization problem. They are now included in the calculation of the mean overall customer satisfaction and the least overall satisfaction of a customer served on route $r \in R$. The complete notation required for the reformulation of MFVRP models is shown in Table 6.

With this preparation done the MFVRP-TC formulates as a set partitioning problem as follows:

$$\min \rightarrow \sum_{r \in R} VC_r \cdot x_r \quad (33)$$

$$\text{s.t. } \sum_{j \in R} a_r^i \cdot x_r = 1 \quad \forall i \in N \quad (34)$$

$$\sum_{r \in R} b_r^t \cdot x_r \leq v_t \quad \forall t \in T \quad (35)$$

$$x_r \in \{0, 1\} \quad j \in R \quad (36)$$

473 Function (33) minimizes the total operational cost of the carrier and is equiv-
 474 alent to (16). Constraints (34) ensure that each customer is included exactly
 475 once in the selected routes. Constraints (35) ensure that no more vehicles of
 476 a given type are used than are available. Finally, constraints (36) define the
 477 domain of the decision variables.

478 The MFVRP-TE is reformulated analogously, with only the objective
 479 function being replaced.

$$\min \rightarrow \sum_{r \in R} VE_r \cdot x_r \quad \text{s.t. } (34)-(36). \quad (37)$$

480 The set partitioning formulations of MFVRP-MS and MFVRP-LS pur-
 481 suing customer satisfaction are straightforward as well.

$$\max \rightarrow 1/n \cdot \sum_{r \in R} MS_r \cdot x_r \quad \text{s.t. } (34)-(36) \quad (38)$$

$$\max \rightarrow LS \quad \text{s.t. } (34)-(36) \quad (39)$$

$$LS \leq LS_r \quad \forall r \in R \quad (40)$$

482 (38) calculates the mean customer satisfaction. The mean customer satis-
 483 faction averages the routes' total satisfaction levels MS_r (i.e. the sum of
 484 relative satisfaction levels of all customers served on route r) by dividing it
 485 by the total number of served customers. Thus, (20) and (38) are equivalent.
 486 Likewise, (39) maximizes of the least overall customer satisfaction.

487 5. Computational experiments

488 Computational experiments are conducted with the models defined in the
 489 previous subsection to investigate the suitability of customer-centric objec-
 490 tives. We examine the extent to which customer preferences influence the
 491 structure of vehicle routes (subsection 5.2), whether these routes are eco-
 492 nomically and ecologically viable (subsection 5.3), and whether customers
 493 benefit equally from this, or not (subsection 5.4). To this end, we analyze
 494 the interaction between vehicle capabilities, expressed as a fleet's degree of
 495 electrification, and customer preferences that can diverge to varying degrees.

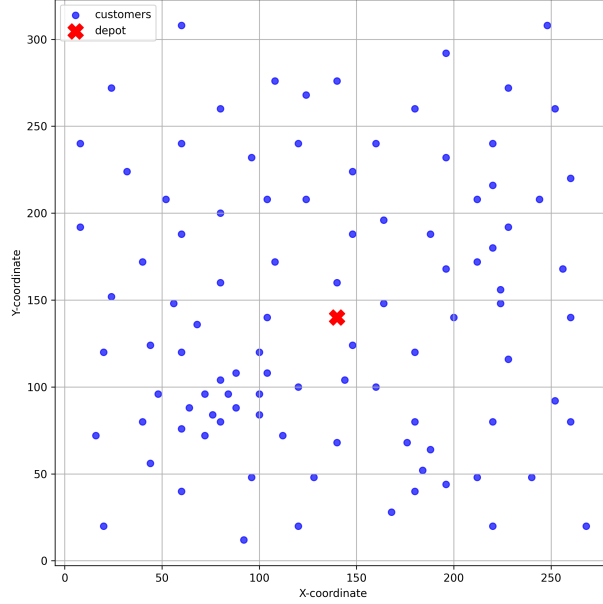


Figure 2: Sketch of Solomons' VRP instance R101.

5.1. Experimental design

The computational study is based on the set of feasible vehicle routes in a MFVRP instance with 100 customers. Considering and solving a single case is sufficient for our purpose, as we are not examining the performance of an optimization method, but rather looking at customer satisfaction under different VRP objectives in principle. In order to study a relevant case of sufficient complexity, we select instance R101 from Solomon's well-known VRP benchmark suite. It provides Euclidean coordinates for 100 customers uniformly distributed around a central depot, see Figure 2. To obtain a long-haul freight transportation setting the coordinates are scaled by a factor of four. The standard customer demand is set at one TEU in each case.

The carriers' fleet consists of ActrosL and eActros600 trucks as examined in subsection 3.2. Regardless of type, all trucks have a capacity of 2 TEU. The total fleet size is set to 50 trucks which is sufficient to serve $n = 100$ customers without employing trucks more than once per period. The fleet mix is defined by the fleet's degree of electrification. I.e., the share of eActros600 trucks in the fleet is set to 20% ($v_e = 10$), 50% ($v_e = 25$), or 80% ($v_e = 40$). Further parameters used in the experiments are shown in Table 7.

Table 7: Relevant vehicle parameters taken from Table 1 and Table 2.

Parameter	c-truck	e-truck
Payload capacity Q	2 TEU	2 TEU
Vehicle cost rate per day c_{day}	€314.24	€375.69
Vehicle cost rate per kilometer c_{km}	€0.57	€0.75
Emission rate per kilometer e_{km}	1.07 kg CO ₂ e	0.44 kg CO ₂ e

Customers are either cost- or emission-sensitive. In order to calculate customer satisfaction levels we first determine the individual ranges for the allocation of cost and emission to each customer, as described in 3.3. Customer preferences are randomly assigned to the 100 customers such that the proportion of emission-sensitive customers ranges from 0% to 100% in steps of 5% creating 21 levels of emission-sensitivity shares. To smooth computational results, 100 combinations of emission-sensitive customers are randomly chosen for each level. Thus, a total of $3 \cdot 21 \cdot 100 = 6.300$ MFVRP instances are created. Each instance is solved with minimizing of total cost (MFVRP-TC) and total emission (MFVRP-TE) as well as maximizing mean and least overall customer satisfaction, i.e. MFVRP-MS and MFVRP-LS, respectively.

Table 8: Pre-processed records for the set partitioning formulation of the MFVRP.

R	a_r^1	a_r^2	$\dots$	a_r^i	$\dots$	a_r^n	b_r^c	b_r^e	VC_r	VE_r	MS_r	LS_r
$\vdots$												
r	0	1	...	1	...	0	1	0	€409.53	178kg	67%	44%
$r + 1$	0	1	...	1	...	0	0	1	€501.08	74kg	69%	41%
$\vdots$												

All instances are solved using the set partitioning formulation outlined in subsection 4.3 which requires a comprehensive pre-processing. At first, route set R is generated for the R101 benchmark under the dual load constraint. It consists of $\frac{n!}{(n-2)!} = 9.900$ different feasible vehicle routes. For each route, four performance measures are computed for each route, namely the total operational vehicle cost VC , the total GHG emission VE measured on a WTW basis, and the mean overall customer satisfaction MS as well as the least overall satisfaction LS observed for customers served on the route. Note

533 that each pair of customers can be served by each of the two truck types.
534 Table 8 shows the records taken for a sample route which is executed by
535 either types of trucks. This pre-processing provides all data necessary to
536 solve the 6.300 MFVRP instances under the four objective functions. The
537 models have been implemented in `Python 3.13.5`, using `Gurobi 11.0.0` as
538 solver.

539 5.2. Structure of vehicle routes

540 The illustrative example in subsection 3.4 suggests that when maximizing av-
541 erage satisfaction optimal solutions may differ from those obtained by min-
542 imizing total cost or total emission. To explore the effect, we count the
543 number of customer pairs (routes) in the optimal solutions of each MFVRP-
544 MS instance that are not part of the optimal solutions of the corresponding
545 MFVRP-TC or MFVRP-TE instances. The results are shown in Figure 3(a-
546 c) for different degrees of electrification of the fleet. In the diagrams, the
547 ordinate shows the percentage of customer pairs appearing in MFVRP-MS
548 solutions but not in solutions of the other two models. Higher percentages
549 indicate greater structural differences between the solutions being compared.
550 The following observations are taken:

- 551 • With a low electrification rate of 20% (see Figure 3a), approximately
552 55% and 45% of all customer pairs in optimal MFVRP-MS solutions
553 are not included in the minimum-cost and minimum-emission solutions,
554 respectively, when the share of emission-sensitive customers is 0%. The
555 differences increase in both cases with increasing share of emission-
556 sensitive customers up to a maximum of about 70%. The maximum
557 is reached when the share of emission-sensitive customers is 20%. If
558 this proportion continues to increase, the observed deviations decrease
559 for both models. Interestingly, more than 95% of the customer pairs
560 selected by the MFVRP-MS model also appear in the corresponding
561 minimum-cost solutions where all customers are emission-sensitive.
- 562 • With an electrification rate of 50% (see Figure 3b), optimal MFVRP-
563 MS solutions differ by almost 90% from minimum-cost solutions when
564 all customers are cost-sensitive. At the same time, the MFVRP-MS se-
565 lects almost the same customer pairs that minimum-emission solutions
566 do. With an increasing proportion of emission-sensitive customers,
567 the difference to minimum-emission solutions increases sharply and at-
568 tains a maximum of 85% when the proportion of emission-sensitive

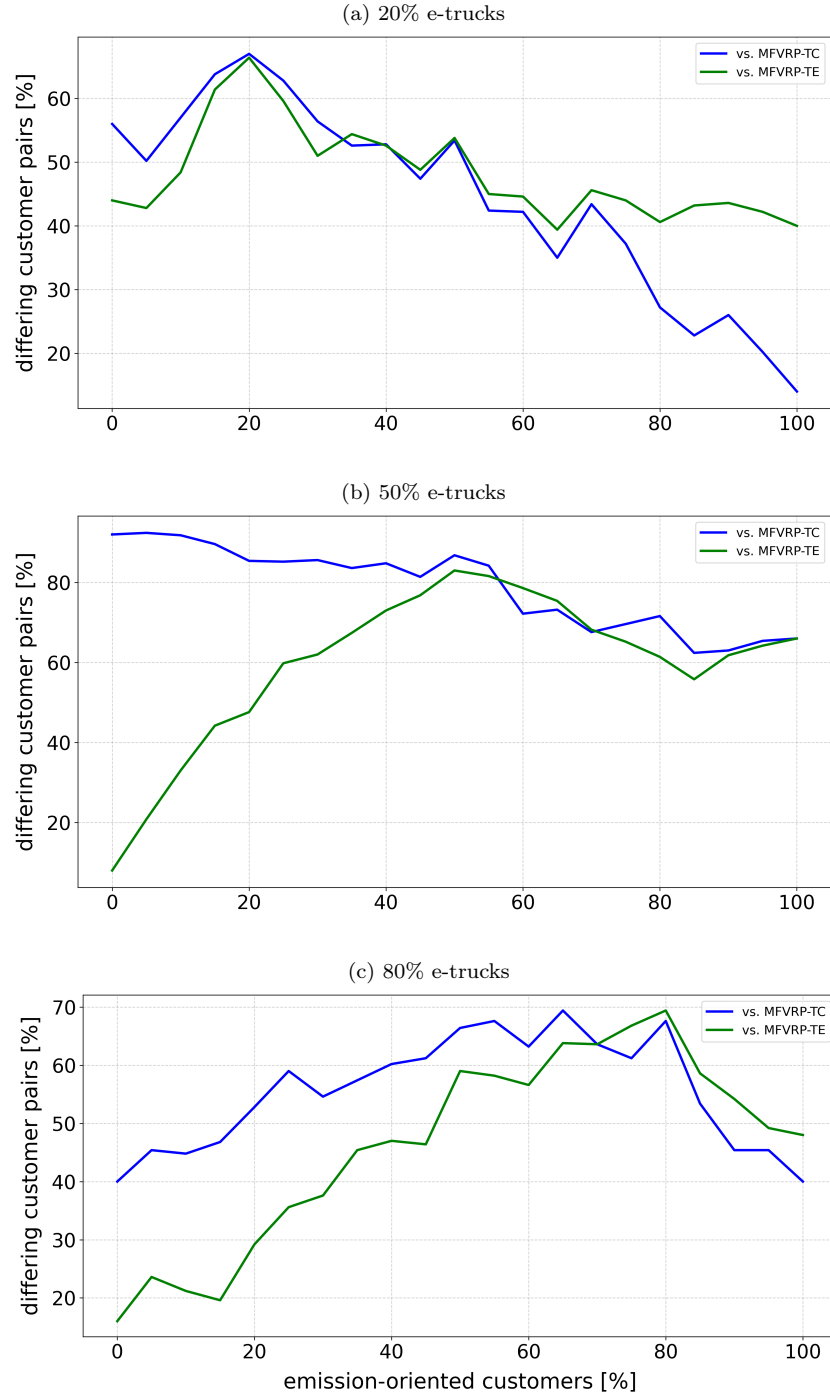


Figure 3: Structural differences of MFVRP-MS solutions compared with MFVRP-TC solutions (blue) and MFVRP-TE solutions (green) for different degrees of electrification.

customers reaches 50%. The observed deviation against the MFVRP-MS decreases for both models as the proportion of emission-sensitive customers increases.

- With a high electrification rate of 80%, MFVRP-MS solutions are very similar to minimum-emission solutions while customers are cost-sensitive, see Figure 3c. The difference to minimum-cost solutions is somewhat larger. As the proportion of emission-sensitive customers increases, the similarity of MFVRP-MS solutions with both, minimum-cost and minimum-emission solutions decreases. More than 60% of the selected customer pairs differ from each other, while 65% to 80% of the customers are emission-sensitive. For even higher percentages the similarity of solutions increases again.

Looking at the plots in Figure 3 shows that minimum-emission solutions closely resemble MFVRP-MS solutions when no or only few customers are emission-sensitive. Satisfaction of a cost-sensitive customer i is assessed as the relative position of its allocated cost c_i within the range $\{\underline{c}_i, \bar{c}_i\}$, rather than the absolute cost reduction. Customers located far away from the depot typically face larger cost ranges due to higher upper cost limits, whereas those closer to the depot experience smaller ranges and are thus more affected by a given cost change. Hence, e-trucks are predominantly assigned to customers for whom the relative impact is lower, i.e. those ones located in the periphery of the depot. At the same time, serving these customers by emission-intensive routes is acceptable as they are indifferent to emissions declarations. As a result, the tour plan becomes more eco-friendly, even though minimizing total emission is not explicitly pursued. As the share of emission-sensitive customers increases, emission reductions become more costly, such that MFVRP-MS solutions gradually approach minimum-cost solutions.

In the MFVRP-MS, the coexistence of cost- and emission-sensitive customers amplifies the underlying trade-offs, resulting in the formation of new route structures which attempt to balance the cost and emission objectives. If the shares of cost- and emission-sensitive customers are similar, the MFVRP-MS selects many customer pairs not generated by the other models. Here, the conflict between economic and environmental goals is most pronounced. Furthermore, it can be seen that the allocation of e-trucks loses its discriminatory impact when they dominate the fleet. With an increasing proportion of emission-sensitive customers, this effect intensifies.

606 The solutions of MFVRP-MS and the other models differ the most, when
607 customer preferences are very heterogeneous. In this area, cost and emission
608 objectives directly interact with each other, prompting the satisfaction-based
609 model to produce compromise solutions. In contrast, the models converge
610 when customer preferences are quite homogeneous, i.e. the proportion of
611 emission-sensitive customers is either very low or very high. This is because,
612 the preference distribution and the technological configuration of the fleet
613 narrow the solution space.

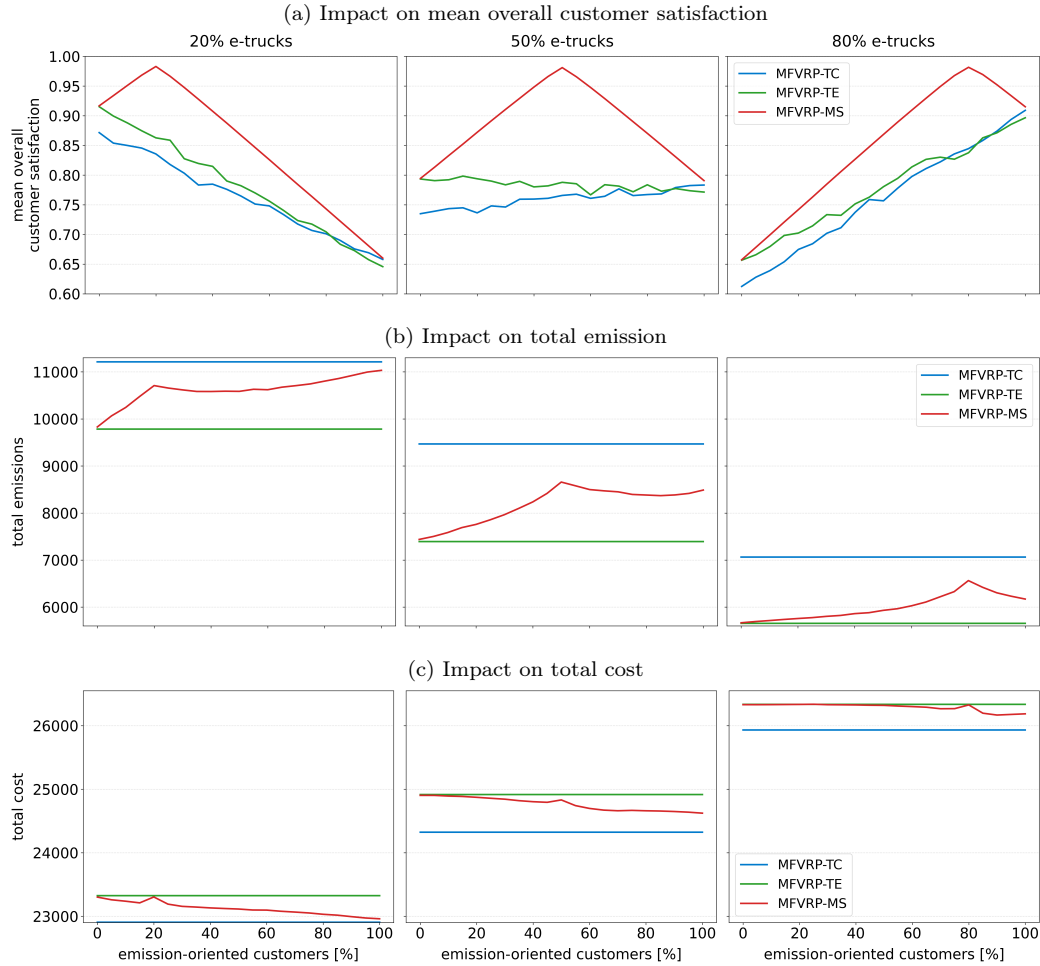


Figure 4: Customer satisfaction, total cost, and total emission of solutions generated by the MFVRP models under heterogeneous customer preferences and varying degrees of fleet electrification.

614 5.3. Economical and ecological performance of vehicle routes

615 Having observed substantial structural differences between the solutions ob-
616 tained by maximizing mean overall customer satisfaction as well as total cost
617 or emission, the performance measures of the different solutions are analyzed.
618 Figure 4 shows the performance measures produced by the three models, rep-
619 resented by blue, green, and red lines. Three subfigures show the impact of
620 the selected objective function on mean overall customer satisfaction (4a),
621 total emission (4b), and total cost (4c). The following observations are taken
622 from Figure 4a:

- 623 • The highest mean overall customer satisfaction is achieved when the
624 fleet mix corresponds to the proportion of emission-sensitive customers,
625 enabling a carrier to respond perfectly to demand. Then, up to 95%
626 mean overall customer satisfaction is reachable if this target is pursued.
627 When the proportion of emission-sensitive customers reduces, overall
628 customer satisfaction declines continuously. The same applies when the
629 proportion exceeds the fleets' electrification degree.
- 630 • When there are no or just a few emission-sensitive customers and the
631 carrier minimizes total emission, mean overall customer satisfaction is
632 always slightly higher than if the carrier minimizes total cost. This is
633 independent of the fleet mix. On the other hand, when all customers
634 are emission-sensitive and the carrier minimizes cost, mean overall cus-
635 tomer satisfaction is always greater than when the carrier minimizes
636 total emission. This seemingly counter-intuitive result was already ap-
637 parent in the structural analysis of subsection 5.2.
- 638 • When the electrification rate of the fleet is low (20%) and the carrier
639 minimizes total cost or emission, customer satisfaction declines as the
640 proportion of emission-sensitive customers increases. Conversely, when
641 the electrification rate is high (80%) and the carrier minimizes total
642 cost or emission, customer satisfaction increases as the proportion of
643 emission-sensitive customers increases. Finally, when the electrification
644 rate is balanced (50%) and the carrier minimizes total emission, mean
645 overall customer satisfaction is almost independent of the proportion
646 of emission-sensitive customers. If, on the other hand, the carrier mini-
647 mizes total cost, mean overall customer satisfaction appears to increase
648 slightly as the number of emission-sensitive customers rises.

Figure 4b and 4c show the effects of the carrier’s route planning on emission and cost. It is clear that both, total emission and total cost are independent of the proportion of emission-sensitive customers when either total emission or total cost is minimized. When the carrier maximizes customer satisfaction, it is also clear that total emission and total cost are above the corresponding minimum values. More importantly, they are always below the total emission achieved when total cost is minimized and below the total cost achieved when total emission is minimized. In other words, the trade-off between cost and emission achieved by maximizing mean overall customer satisfaction is in no way doubly dominated. It also appears that maximizing customer satisfaction goes hand in hand with minimizing total emission when there are no or only few emission-sensitive customers.

The comparison of the models demonstrates that the MFVRP-MS strikes a balance between cost, emission, and customer satisfaction. The characteristic inverted U-shape of the red curves in the diagrams shows that the model achieves highest benefit when customer preferences are heterogeneous. Coexistence between cost- and emission-sensitive customers works best when their preferences reflect the fleets’ degree of electrification. As the fleet becomes more electrified, the absolute differences in cost and emission between the MFVRP-MS and the other models decrease. Clearly, the MFVRP-MS not just pursues a weighted mix of cost and emission minimization, it actually creates a balance between economic and environmental goals.

5.4. *Maximizing customer satisfaction for the most dissatisfied customer*

Having considered the carrier’s goal of maximizing mean overall customer satisfaction, we now turn to the perspective of customers who still might be dissatisfied with the routes planned for them. As outlined in 3.3, the question addresses the fairness perceived among the customers served in a process. This study analyzes the relationship between the mean and the least customer satisfaction achieved by the MFVRP-MS and MFVRP-LS models. The experiments evaluated refer to a balanced (50%) electrification rate of the fleet. Different rates lead in turn to similar findings.

Recall from 4.2 that the MFVRP-LS is more difficult to solve than the MFVRP-MS due to the min-max objective. To linearize the function, an additional variable and coupling constraints are required, which increases the number of active constraints at optimality. The rise in combinatorial complexity generally leads to longer computation times. In our experiments they scaled up by a factor of 5.3 for instances with 100 customers.

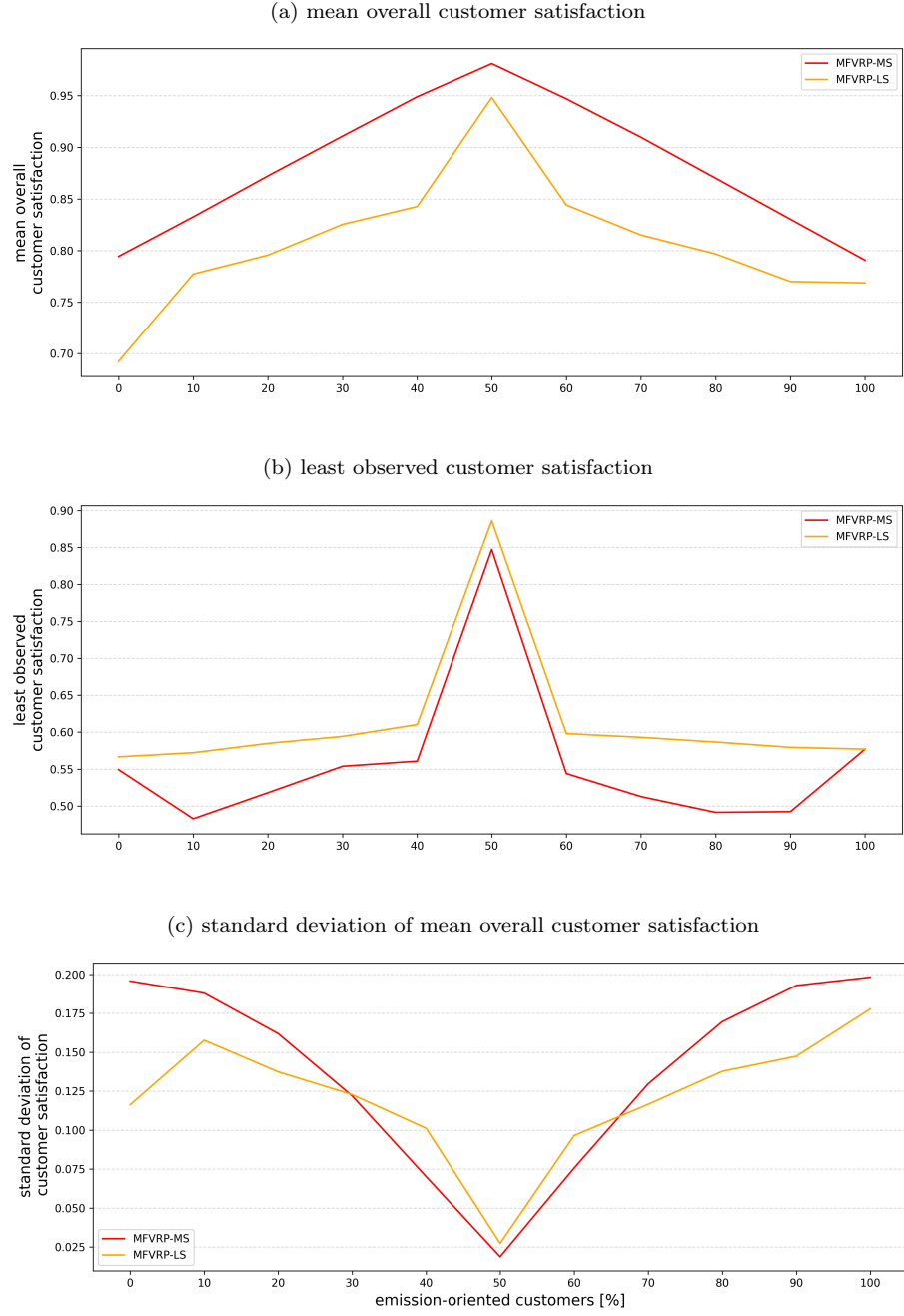


Figure 5: Comparison of MFVRP-MS and MFVRP-LS solutions for heterogeneous customer preferences and a 50% degree of fleet electrification.

Figure 5 shows three performance measures describing the fulfillment of customer preferences by the two competing models. Figure 5a indicates the mean overall customer satisfaction as defined above. Figure 5b indicates the satisfaction of the customer least satisfied with the produced tour plan. Finally, Figure 5c indicates the mean standard deviation of customer satisfaction, determined as the average of the standard deviations over all instances. We are interested in this measure because treating customers fairly and equally can be looked at as enabling similar levels of individual satisfaction for all customers. In other words, a fair tour plan ensures a low standard deviation in customer satisfaction while maintaining a high average level of satisfaction.

In Figure 5a, the red line corresponds to the red line in the left diagram of Figure 4b. The orange line in Figure 5a represents the mean overall customer satisfaction that is achieved when the least customer satisfaction is maximized. Clearly, this line has to fall below the blue line, which happens in our calculations with an almost constant distance. Conversely, the least satisfaction of a customer, which is observed when mean overall customer satisfaction is maximized, falls below the corresponding measure when the satisfaction of the least satisfied customer is maximized. This is verified in Figure 5b. Interestingly the distance between the red and the orange line is much smaller now. It takes the smallest values (below of 1%) when no, all, or around one half of the customers are emission-sensitive. At these extremes, there is hardly no difference between employing the MFVRP-MS and the MFVRP-LS except of the computation times.

Figure 5c shows the standard deviation of customer satisfaction generated by the MFVRP-MS (red line) and the MFVRP-LS (orange line). The former model appears to be slightly fairer in terms of standard deviation than the latter model when the proportion of emission-sensitive customers is between 30% and 70%, i.e. when customer preferences are very heterogeneous. When customer preferences are more homogeneous, the MFVRP-LS provides solutions that appear to be slightly fairer. This pattern illustrates that heterogeneity of customer preferences fosters not only higher mean overall customer satisfaction but also more equitable outcomes. Both models balance customers with different preferences by assigning e-trucks and c-trucks in a way that extreme dissatisfaction is mitigated. Heterogeneity acts as a stabilizing mechanism within the optimization environment. When preferences are more homogeneous, the mechanism loses its capability and the models revert to solutions that privilege a single direction of optimization.

724 To summarize, there is no dominant model for creating fair tour plans.
725 In cases where the MFVRP-LS generates fairer solutions, the difference to
726 MFVRP-MS solutions is only minor. We therefore believe that it is not worth
727 the effort to employ the more complex MFVRP-LS model.

728 6. Managerial insights and outlook

729 This paper investigates customer satisfaction as a new operational target
730 for carriers competing in road freight transportation markets that currently
731 undergo a transition. An upcoming demand for clean transport services is
732 putting increasing pressure on transport companies to gradually convert their
733 fleets to electric vehicles. In response, the paper introduces the Mixed Fleet
734 Vehicle Routing Problem with consideration of the fleet’s degree of electrifi-
735 cation and individual preferences of the customers. A concept for evaluating
736 customer satisfaction with the carrier’s decisions making is proposed and
737 embedded in the optimization problem.

738 The tests conducted to compare different objectives functions are based on
739 profound cost and emission calculations for conventional and electric vehicles.
740 It turned out that tour plans generated with the new model remain consistent
741 with established approaches minimizing total transport cost or emissions
742 as long as customer preferences are fairly homogeneous. However, when
743 customer preferences exhibit heterogeneity, the fleet mix plays an important
744 role. It is then crucial that the proportion of eco-friendly vehicles reflects
745 more or less the proportion of emission-sensitive customers. To ensure this,
746 the carrier should gradually adapt its fleet in the long term. In the short term,
747 it can try to achieve a suitable match between the fleet mix and customer
748 preferences by deciding which transport orders to accept. When a carrier
749 has neither the ability to adjust the composition of its fleet nor to reject
750 customers, it is recommended to minimize the distance traveled by the fleet
751 (and thus total emission) as this generally has a greater impact on customer
752 satisfaction than minimizing total cost.

753 While the current application was restricted to a stylized setting with
754 a predefined fleet mix and homogeneous dual load customer demand, the
755 underlying framework can be generalized. An extension towards general less-
756 than-truckload services is possible. Furthermore, the model can be expanded
757 to include further customer attitudes, such as indifferent or mixed-preference
758 customers, thereby enabling an even more nuanced depiction of real-world
759 heterogeneity. Overall, the research illustrates that customer-centric and

760 sustainability-aware VRPs not only improve the representational accuracy
 761 of logistics models but also offer relevant managerial guidance in navigating
 762 the ongoing transition of transport systems.

763 7. Assignment problem perspective

764 Let the customer nodes be split into two disjoint subsets N_1, N_2 such that
 765 $N_1 \cup N_2 = N$ and $N_1 \cap N_2 = \emptyset$. Any pair of customers $(i, j, t) \in N_1 \times N_2 \times T$
 766 constitutes a dual load tour with vehicle type $t \in T$ of length $TL(i, j) =$
 767 $d_{0,i} + d_{i,j} + d_{j,0}$. For total cost and emission declaration of tour/assignment
 768 (i, j, t) holds

$$\begin{aligned} TC(i, j, t) &= c_t^{fix} + c_t^{var} \cdot TL(i, j) \\ TE(i, j, t) &= e_t^{fix} + e_t^{var} \cdot TL(i, j) \end{aligned}$$

where $TL(i, j)$ is the total tour length as defined before. Based on total cost and emissions of the tour, individual preference values OS_i^t and OS_j^t can be calculated for each vehicle type t and customer. For tour/assignment (i, j, t) , the total preference contribution is, thus,

$$OS(i, j, t) = OS_i^t + OS_j^t$$

769 . An assignment problem for maximizing overall satisfaction analogue to
 770 MFVRP-MS can be formulated as

$$\max \rightarrow \sum_{(i,j,t)} x_{(i,j,t)} \cdot OS(i, j, t) \quad (41)$$

$$\sum_{i \in N_1, t \in T} x_{(i,j,t)} = 1 \quad \forall j \in N_2 \quad (42)$$

$$\sum_{j \in N_2, t \in T} x_{(i,j,t)} = 1 \quad \forall i \in N_1 \quad (43)$$

$$\sum_{i \in N_1, j \in N_2} x_{(i,j,t)} \leq v_t \quad \forall t \in T \quad (44)$$

$$x_{i,j,k} \in \{0, 1\} \quad \forall i \in N_1, j \in N_2, t \in T \quad (45)$$

771 Thus, the routing problem can be formulated as matching problem in the
 772 bipartite graph with node sets N_1 and N_2 and multiple arcs connecting any

773 pair of nodes. Thereby, each vehicle type $t \in T$ refers to one arc between
774 node pair (i, j) .

775 When there are no capacity restrictions on vehicle types t , the set of
776 arcs for node pair (i, j) can be reduced to the optimal vehicle type t^* with
777 $t^* = \arg \max_{t \in T} OS(i, j, t)$. In this case, the optimal assignment problem
778 can be solved to optimality by the Hungarian algorithm in polynomial time.
779 Similar results follow for classic total objectives like minimizing total cost,
780 distance or emissions.

781 For maximizing the least satisfied customer, the MILP is to be adjusted
782 slightly. Let $\underline{OS}(i, j, t) = \min(OS_i^t, OS_j^t)$ define the more unsatisfied cus-
783 tomer on tour (i, j, t) . Then, the corresponding assignment problem can be
784 formulated as

$$\max \rightarrow LS \quad \quad \quad s.t. (42) - (45) \quad (46)$$

$$x_{(i,j,t)} \cdot \underline{OS}(i, j, t) \leq LS \quad \forall i \in N_1, j \in N_2, t \in T \quad (47)$$

785 **Acknowledgements:** This research is funded by the German Research
786 Foundation (DFG) under grant 567015469.

787 Appendix A. Example Appendix Section

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